Homework #4

Math 104: Intro to Analysis

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Contents

0 Various Theorems

1 Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : (X, d_X) \to (Y, d_Y)$ is called *Lipschitz continuous* if there is a constant $L > 0$ so that

$$d_Y(f(p), f(q)) \leq L d_X(p, q)$$

for all $p, q \in X$. For $\alpha > 0$, we say f is *Hölder continuous (with exponent α)* if there is a constant $C > 0$ so that

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$$

for all $p, q \in X$. In particular, Hölder continuous with exponent 1 means the same thing as Lipschitz.

(a) Fix a point $p \in X$. Show that the map $g : X \to \mathbb{R}$ defined by $g(x) = d(x, p)$ is Lipschitz continuous, with constant $L = 1$.

(b) Prove that any Hölder continuous function (with any exponent $\alpha > 0$) is uniformly continuous.

(c) Prove that if f is Lipschitz continuous and bounded, then f is Hölder continuous with exponent α, for any $0 < \alpha \leq 1$.

(d) Let $[a, b]$ be a closed interval, and suppose $f : [a, b] \to \mathbb{R}$ is C^1 (meaning its first derivative exists and is continuous on $[a, b]$). Show that f is Lipschitz continuous.

(e) Suppose $f : \mathbb{R} \to \mathbb{R}$ is Hölder continuous with exponent $\alpha > 1$. Show that f is constant.

2 Problem 2

On the interval $(100, \infty)$, define functions $f(x) = x + \sin(x)$, $g(x) = x$, $p(x) = x + \sin(x)\cos(x)$, and $q(x) = p(x)e^{\sin(x)}$.

(a) Using facts we proved about limits and derivatives, prove that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1, \quad \lim_{x\to\infty} \frac{f'(x)}{g'(x)} \text{ DNE}, \quad \lim_{x\to\infty} \frac{p(x)}{q(x)} \text{ DNE, and } \lim_{x\to\infty} \frac{p'(x)}{q'(x)} = 0.$$

(b) For both scenarios $\frac{f(x)}{g(x)}$ and $\frac{p(x)}{q(x)}$, explain why these results do not contradict L'Hôpital's rule.

Note: even though we haven't learned about sin *and* cos *yet, you are free to use any familiar properties of them you wish.*